

Disk Quotas

- one user of a multi-user system can use all the disk space
- to manage this, Linux supports *disk quotas*
- you can enforce how much disk space individuals/groups can use

Enabling Quota Support

quota

display disk usage and limits
installs other utility packages

Configuration

modify */etc/fstab* for any partitions you want quota support
options

<code>usrquota</code>	use user quota
<code>grpquota</code>	use group quota

example line in */etc/fstab*:

```
/dev/nvme0n1p4 / ext4 usrquota,grpquota 1 1
```

- activates user and group quota for */dev/mvme0n1p4* partition
- partition is mounted at */*
- to activate root can use **quotaon** command
- to turn off **quotaoff**

quotacheck

creates and checks quota files:

1. *aquota.user*
2. *aquota.group*

installed by `quota` package

<code>c</code>	create files
<code>u</code>	check user quota
<code>g</code>	check group quota

example

```
quotacheck -u /home
```

creates the quota file in the */home* directory

Setting User Quotas

edquota

- edit user quotas
- uses a text editor which a temporary configuration file **/etc/quotatab**
- when you exit the utility, it writes the temporary file to the kernel

example

`edquota sally` shows:

```
Disk quotas for user sally (uid 21810):
Filesystem    blocks      soft      hard    inodes    soft      hard
/dev/sdc4      97104    1048576    1048576    1242         0         0
```

both **blocks** and **inodes** have **soft** and **hard** limits

blocks	number of blocks being used block size is configured by the filesystem
inodes	number of files being used
soft	issue warnings when exceeded have a grace time limit - if soft quota is exceeded for greater than grace limit you can't create any new files
hard	max limit
-t	sets the grace period for a filesystem
-g	edit group quotas
-u	edit user quotas